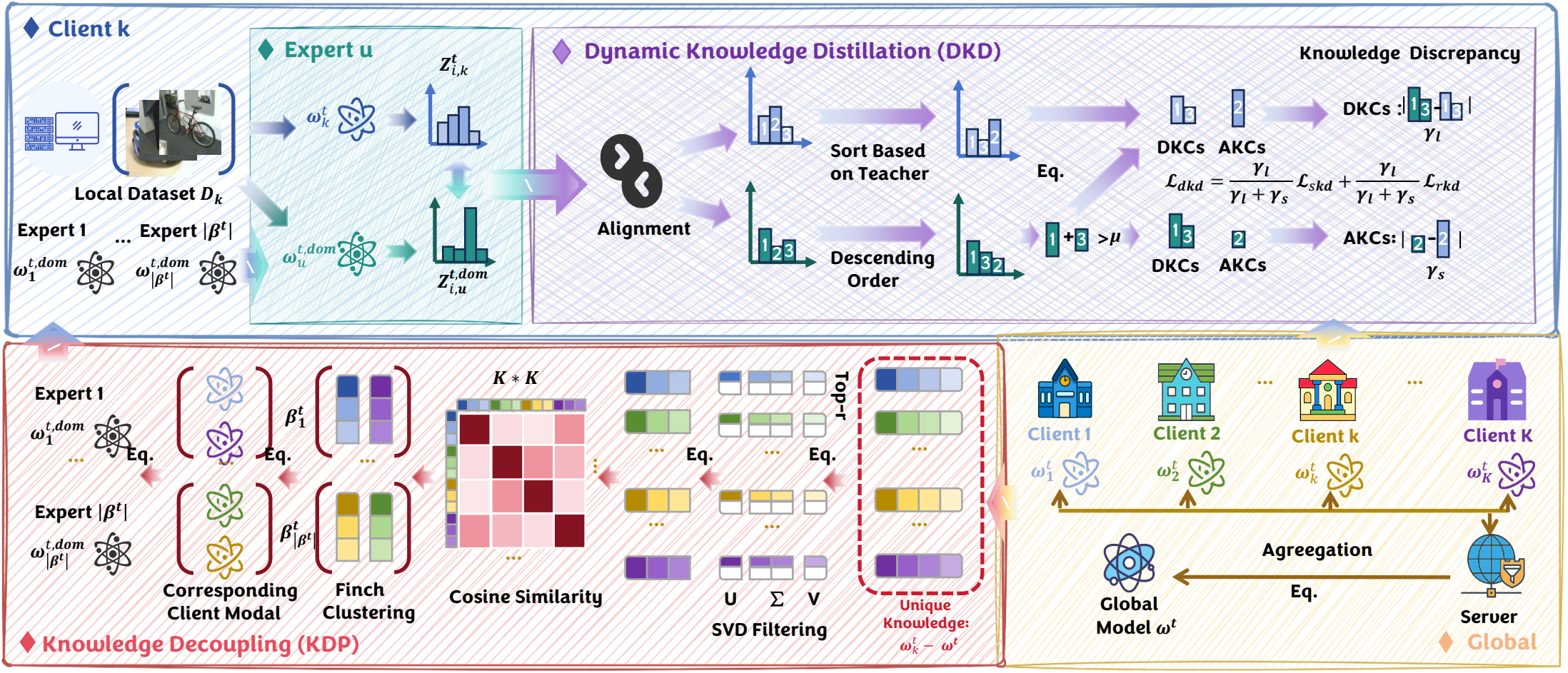
Client k
Local Dataset D_k
Expert 1
Expert $|\beta^t|$
$\omega_1^{t,dom}$
$\omega_{|\beta^t|}^{t,dom}$
Expert u
ω_k^t
$\omega_u^{t,dom}$
$Z_{i,k}^t$
$Z_{i,u}^{t,dom}$
Dynamic Knowledge Distillation (DKD)
Alignment
Sort Based on Teacher
Descending Order
Eq.
$1+3 > \mu$
Knowledge Discrepancy
DKCs
AKCs
DKCs :
AKCs:
γ_l
γ_s
$\mathcal{L}_{dkd} = \frac{\gamma_l}{\gamma_l+\gamma_s}\mathcal{L}_{skd} + \frac{\gamma_l}{\gamma_l+\gamma_s}\mathcal{L}_{rkd}$
Expert 1
Expert $|\beta^t|$
$\omega_1^{t,dom}$
$\omega_{|\beta^t|}^{t,dom}$
Eq.
β_1^t
$\beta_{|\beta^t|}^t$
Corresponding Client Modal
Finch Clustering
$K * K$
Cosine Similarity
Top-r
U Σ V
SVD Filtering
Unique Knowledge:
$\omega_k^t - \omega^t$
Knowledge Decoupling (KDP)
Client 1
Client 2
Client k
Client K
ω_1^t
ω_2^t
ω_k^t
ω_K^t
Agreegation
Eq.
Global Model ω^t
Server
Global